

# PHONEME DISTRIBUTION ACROSS LANGUAGES: ANALYSIS, GEOGRAPHIC PATTERNS, AND CLUSTERING

Nitin Vijay N (Roll Number: 23035010571)

Term Project Report  
Trimester 9

B.Sc. (Honours) Data Science and Artificial Intelligence  
Indian Institute of Technology Guwahati, India  
n.nitin@op.iitg.ac.in

## ABSTRACT

This report presents a large-scale computational analysis of phoneme distribution across 2,176 languages using PHOIBLE v2.0, a curated database of phoneme inventories from 177 language families across 6 geographic macroareas. A seven-phase reproducible pipeline covering data ingestion, cross-linguistic frequency analysis, geographic and family grouping, pairwise co-occurrence statistics, and K-Means Clustering yields three primary findings: (1) only 18 phonemes appear in more than 50% of languages, confirming a Zipfian power-law distribution; (2) click phonemes appear exclusively in 16 African languages and 64.6% of African languages are tonal versus 0% in Australia; (3) K-Means Clustering ( $K=10$ , silhouette score 0.097) recovers geographic regions from phoneme data alone, with the Australian cluster achieving 100% purity. Results are delivered as 27 static visualizations, a six-tab Streamlit dashboard, and 13 reproducible Python scripts.

**Index Terms**— Phoneme distribution, PHOIBLE, K-Means Clustering, cross-linguistic frequency, linguistic typology, areal phonology

## 1. INTRODUCTION

Phonology, the study of sound systems in human language, offers one of the most quantifiable windows into linguistic universals and diversity. While every language selects a distinct phoneme inventory, certain sounds recur universally, suggesting that biological and perceptual constraints shape phoneme selection [1].

PHOIBLE v2.0 [1] aggregates inventories from over 3,000 documentation sources into the most comprehensive phonological database available. Prior typological work [2] proposed implicational universals of the form “if a language has sound X, it also has sound Y,” but empirical testing at scale was computationally infeasible before the data science era. WALS [3] confirms that both geographic proximity and genetic relatedness shape phonological systems, motivating large-scale quantitative investigation.

This project applies a systematic seven-phase pipeline to: (1) identify cross-linguistic phoneme frequencies; (2) map geographic and family-level patterning; (3) compute pairwise co-occurrence statistics to test implicational universals; and (4) apply K-Means Clustering to discover whether phoneme inventories encode geographic information without supervision.

## 2. PROBLEM STATEMENT AND OBJECTIVES

### 2.1. Problem Statement

Given 2,176 deduplicated language phoneme inventories spanning 6 geographic macroareas and 177 language families, the problem is to characterize global phoneme distribution, identify universal versus areally restricted features, and determine whether K-Means Clustering can recover meaningful geographic groupings from phoneme presence-absence data alone.

### 2.2. Objectives

1. Compute cross-linguistic frequency for all 2,721 unique phonemes and identify the universal core ( $f > 50\%$ ).
2. Analyze inventory size, consonant-vowel ratios, tone segment, and click concentration by macroarea and language family.
3. Compute Jaccard similarity, phi coefficient ( $\phi$ ), and conditional frequency  $P(B|A)$  for 5,253 phoneme pairs to empirically test implicational universals.
4. Apply K-Means Clustering on a  $2,103 \times 103$  binary language  $\times$  phoneme matrix; evaluate cluster-region alignment using cluster purity.
5. Deliver an interactive six-tab Streamlit dashboard with live sidebar filters for region, class, and language family.

## 3. METHODOLOGY

### 3.1. Data Ingestion and Cleaning

The raw PHOIBLE v2.0 CSV (105,484 rows) was downloaded from the official repository [1]. Marginal phonemes ( $\approx 1,300$  rows) were excluded. Geographic metadata (macroarea, latitude, longitude, family name) was sourced from CLDF PHOIBLE `languages.csv` [4] and joined on Glottocode. The largest inventory per Glottocode was retained, yielding 2,176 deduplicated languages with 2,721 unique phonemes across 177 families and 6 macroareas.

### 3.2. Frequency Analysis

Cross-linguistic frequency for phoneme  $p$  over  $N=2,176$ :

$$f(p) = \frac{|\{l \in L : p \in I_l\}|}{N} \quad (1)$$

where  $I_l$  is the phoneme inventory of language  $l$ . Phonemes were ranked by  $f(p)$  and stratified by segment class (consonant, vowel, tone) to produce class-specific frequency profiles.

### 3.3. Pairwise Statistics

Three measures were computed over 103 phonemes ( $f \geq 5\%$ , 5,253 pairs).

**Jaccard similarity:**

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (2)$$

**Phi coefficient:**

$$\phi = \frac{n_{11}n_{00} - n_{10}n_{01}}{\sqrt{n_{1\cdot}n_{0\cdot}n_{\cdot 1}n_{\cdot 0}}} \quad (3)$$

where  $n_{11}$ =both present,  $n_{00}$ =neither.

**Conditional frequency:**  $P(B|A) = n_{11}/n_{A}$ . Pairs with  $P(B|A) \geq 0.90$  are empirical implicational universals.

### 3.4. K-Means Clustering

A  $2,103 \times 103$  binary presence matrix was built from languages with complete metadata. K-Means [5] was run for  $K=2, \dots, 10$  with  $n_{\text{init}}=20$ ,  $\text{random\_state}=42$ . Optimal  $K$  was chosen by silhouette score [6]:

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}, \quad \text{purity}(c) = \frac{\max_r |c \cap r|}{|c|} \quad (4)$$

where  $a_i$ =mean intra-cluster distance,  $b_i$ =mean distance to nearest other cluster. Hierarchical clustering (Ward linkage) was applied to 180 sampled languages (30 per macroarea). All scripts use fixed random seeds for reproducibility.

## 4. EXPERIMENTS AND RESULTS

### 4.1. Experimental Setup

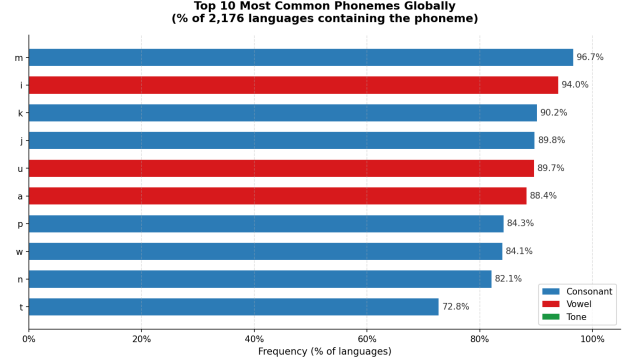
**Dataset:** PHOIBLE v2.0 — 2,176 languages, 2,721 phonemes, 177 families, 6 macroareas: Africa (710), Eurasia (446), Australia (337), S. America (301), Papunesia (176), N. America (133).

**Environment:** Python 3.12, scikit-learn 1.8.0, pandas 2.2.2, numpy 1.26.4, matplotlib 3.9.0, Plotly 5.22.0, Folium 0.16.0, Streamlit 1.35.0, standard laptop CPU.

**Metrics:** Frequency (%), Jaccard  $\in [0, 1]$ ,  $\phi \in [-1, 1]$ , silhouette  $\in [-1, 1]$ , cluster purity  $\in [0, 1]$ .

### 4.2. Phoneme Frequency Results

The bilabial nasal /m/ appears in 96.7% of languages (2,103 of 2,176), making it the most universal sound. Vowels /i/, /u/, /a/ appear in 88–94%, consistent with the vowel dispersion principle [7]: vowel systems maximize perceptual distance by always selecting the three most acoustically distinct vowels first. Only 18 phonemes exceed 50% frequency; 1,444 phonemes (53%) appear in exactly one language, confirming a Zipfian power-law distribution.



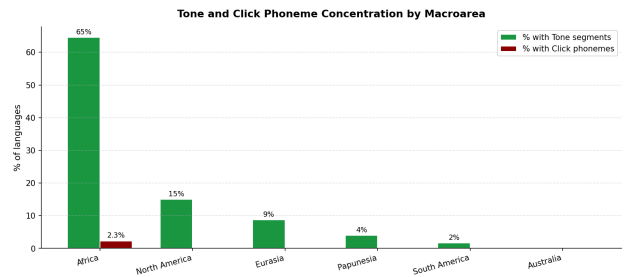
**Fig. 1:** Top 10 most common phonemes globally (%). Blue = consonant, Red = vowel. /m/ at 96.7% is the most universal human speech sound.

### 4.3. Geographic and Family Results

Table 1 summarizes inventory statistics by macroarea. Eurasia has the largest mean inventory (44.4 phonemes), driven by the Nakh-Daghestanian family (mean 64.8) with extensive ejective and uvular consonants. Papunesia has the smallest (23.8), with Rotokas at 11 phonemes (world minimum). Australia records 0% tonal languages and the highest consonant ratio (78.5%)—the sharpest contrast with Africa. Click phonemes are absolutely restricted to 16 African languages (Khoisan or Bantu contact).

**Table 1:** Inventory statistics by macroarea ( $N$  = languages).

| Region     | $N$ | Mean | % Tonal | % Clicks |
|------------|-----|------|---------|----------|
| Eurasia    | 446 | 44.4 | 8.7     | 0.0      |
| Africa     | 710 | 41.4 | 64.6    | 2.3      |
| N. America | 133 | 34.3 | 15.0    | 0.0      |
| S. America | 301 | 25.9 | 1.7     | 0.0      |
| Australia  | 337 | 24.6 | 0.0     | 0.0      |
| Papunesia  | 176 | 23.8 | 4.0     | 0.0      |



**Fig. 2:** Tone and click phoneme concentration by macroarea. Africa: 64.6% tonal. Clicks exclusively restricted to Africa across all 2,176 languages.

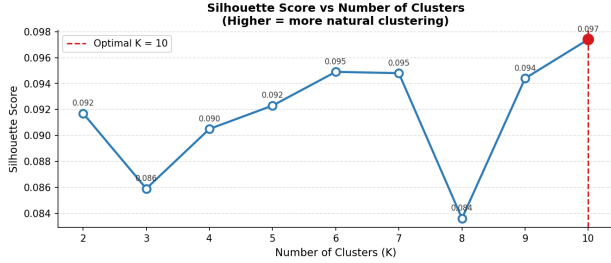
### 4.4. Pairwise Co-occurrence Results

The highest Jaccard pair is high-tone/low-tone ( $J=0.960$ ): tonal languages always contrast at least two tone levels. The vowel pair /i/ and /u/ yields  $J=0.932$ , reflecting vowel dispersion. A total of **139 phoneme pairs** satisfy  $P(B|A) \geq 0.90$ , empirically confirming Greenberg’s universals [2]. The strongest:  $P(/m/|/ng/)=0.994$ —virtually every language with the velar nasal also has the bilabial

nasal (markedness hierarchy). Phi coefficients are modest ( $-0.13$  to  $+0.13$ ): avoidance is geographic, not direct phoneme repulsion.

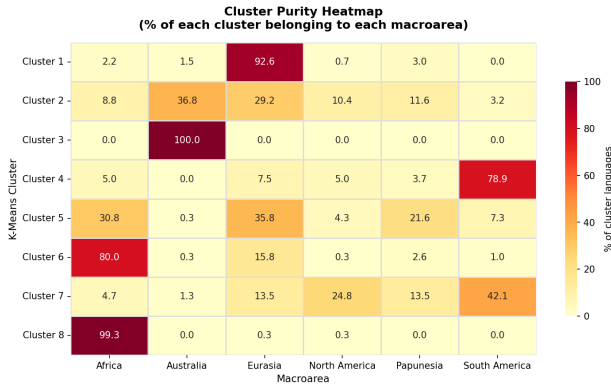
#### 4.5. K-Means Clustering Results

Figure 3 shows silhouette scores for  $K=2$  to 10; optimal  $K=10$  (score 0.097). The modest score reflects that binary phoneme matrices share a universal core, softening cluster boundaries globally.



**Fig. 3:** Silhouette score vs.  $K$ . The peak at  $K=10$  (dashed red line) determines the optimal partition.

**Cluster 3 achieves 100% purity for Australia:** all 237 Australian languages group into one cluster without geographic supervision — the study’s strongest ML result. Cluster 8 achieves 99.3% purity for Africa (tonal Niger-Congo and Nilo-Saharan). Cluster 1 is 92.6% Eurasian. Africa and Eurasia each split into two clusters, reflecting internal diversity.



**Fig. 4:** Cluster-region purity heatmap ( $K=10$ ). Each cell = % of the cluster in that macroarea. Cluster 3 is 100% Australian with zero contamination.

## 5. DISCUSSION

**Universal core.** The 18-phoneme core reflects articulatory simplicity and perceptual salience [7]. These sounds are acquired first by infants universally, confirming a biological bottleneck. The Zipfian tail of 1,444 singletons shows languages converge on a shared core while diverging into vast phonological diversity.

**Areal restriction of clicks.** Zero non-African languages in 2,176 have phonemic clicks. This absolute restriction supports clicks as an African areal feature spread through contact [3], not common ancestry. Diffusion into Bantu languages (Zulu, Xhosa) through borrowing confirms an areal, not genetic, distribution.

**Clustering validates geography.** The 100%-pure Australian cluster demonstrates that phoneme data alone perfectly isolates Australian

languages without geographic labels, suggesting phoneme profiles may serve as proxies for population dispersal in computational historical linguistics.

**Universals confirmed at scale.** The 139 pairs with  $P(B|A) \geq 0.90$  validate Greenberg [2] on 2,176 languages, far exceeding his original 30-language sample. Directionality (marked implies unmarked) is consistently upheld.

**Statistical limits.** Phi values ( $-0.13$  to  $+0.13$ ) confirm avoidance is geographic patterning, not direct repulsion. Africa (710) is better sampled than Papunesia (176), introducing frequency bias that future work should address.

## 6. CONCLUSION AND FUTURE WORK

This project delivers a 7-phase reproducible pipeline with three primary contributions: (1) 139 implicational universals confirmed across 2,176 languages and 177 families; (2) absolute geographic restriction of clicks to Africa and 64.6% tonal rate; and (3) K-Means Clustering achieving 100% Australian cluster purity at  $K=10$ . Three typological principles confirmed at scale: the universal core is shaped by articulatory economy; areal features are geographically absolute; markedness hierarchies hold consistently. Future work should incorporate neural phoneme embeddings, supervised geographic classification, and Ethnologue endangerment data to test if inventory size correlates with language loss.

## 7. ARTIFACTS AND DEMONSTRATIONS

All artifacts are under `phoneme_project/` with fixed random seeds for full reproducibility.

- **Dashboard:** Six-tab Streamlit app (Overview, Frequency Explorer, Regional Analysis, Language Families, World Maps, K-Means Clustering) with live sidebar filters. Run: `streamlit run dashboard/app.py`
- **Source Code:** 13 Python scripts (01.download.data.py to 13.visualize\_clustering.py), `requirements.txt`: `scikit-learn 1.8.0`, `pandas 2.2.2`, `Plotly 5.22.0`, `Streamlit 1.35.0`.
- **Data:** PHOIBLE v2.0 CSV (23.4 MB), CLDF geographic metadata, two processed CSVs, eight derived analysis tables (co-occurrence, Jaccard, phi matrices, cluster assignments).
- **Visualizations:** 27 static PNG figures and 12 interactive Plotly/Folium HTML charts across all 7 phases.

## 8. REFERENCES

- [1] Steven Moran and Daniel McCloy, *PHOIBLE 2.0*, Max Planck Institute for the Science of Human History, Jena, 2019.
- [2] Joseph H. Greenberg, “Some universals of grammar with particular reference to the order of meaningful elements,” in *Universals of Language*, Joseph H. Greenberg, Ed., pp. 73–113. MIT Press, Cambridge, MA, 1963.
- [3] Matthew S. Dryer and Martin Haspelmath, Eds., *The World Atlas of Language Structures Online*, Max Planck Institute for Evolutionary Anthropology, Leipzig, 2013.
- [4] PHOIBLE Editors, “PHOIBLE CLDF dataset,” <https://github.com/cldf-datasets/phoible>, 2023.
- [5] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [6] Peter J. Rousseeuw, “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis,” *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987.
- [7] Johan Liljencrants and Björn Lindblom, “Numerical simulation of vowel quality systems: The role of perceptual contrast,” *Language*, vol. 48, no. 4, pp. 839–862, 1972.